Fractions and Decimals in $\mathbb{Q}$

1. Convert $\frac{3}{8}$ into a decimal.
2. Convert 0.65 into a fraction in simplest form.
3. Convert $-\frac{7}{12}$ into a repeating decimal.
4. Convert -1.375 into a fraction (or mixed number) in simplest form.
5. Convert $0.\overline{45}$ into a fraction in simplest form.
6. Convert $\frac{14}{11}$ into a repeating decimal.
7. A machinist measures a steel pin with a diameter of 0.875 inches. Express this measurement as a fraction in simplest form.
8. A recipe requires $\frac{5}{16}$ of a liter of broth. To use a digital scale that displays decimal values, what is this quantity as a decimal?
9. A stock's price changed by -2.36 dollars over a trading session. Write this change as a simplified mixed number.
10. A sprinter finishes a race in $11\frac{3}{20}$ seconds. Express this total time as a decimal.

Solutions

1. 0.375

   Explanation: Divide 3 by 8 using the bus stop method:

$$8 \,\Big|\, \begin{array}{l} 0.\ 3\ 7\ 5 \\ \hline 3.^{3}0^{6}0^{4}0 \end{array}$$

   Thus, $\dfrac{3}{8} = 0.375$.

2. $\dfrac{13}{20}$

   Explanation: Write the decimal over 100 and divide the numerator and denominator by their greatest common divisor (5):

$$0.65 = \frac{65}{100} = \frac{65 \div 5}{100 \div 5} = \frac{13}{20}.$$

3. $-0.58\overline{3}$

   Explanation: Perform short division (bus stop method) for $7 \div 12$, keeping the negative sign:

$$12 \,\Big|\, \begin{array}{l} 0.\ 5\ \ 8\ 3\ 3\ldots \\ \hline 7.^{7}0^{10}0^{4}0^{4}0\ldots \end{array}$$

   The remainder 4 repeats indefinitely, giving $-\dfrac{7}{12} = -0.58\overline{3}$.

4. $-1\dfrac{3}{8}$ (or $-\dfrac{11}{8}$)

   Explanation: Separate the integer and fractional parts, write 0.375 over 1000, and simplify by dividing by 125:

$$-1.375 = -\left(1 + \frac{375}{1000}\right) = -\left(1 + \frac{375 \div 125}{1000 \div 125}\right) = -1\frac{3}{8} = -\frac{11}{8}.$$

5. $\dfrac{5}{11}$

   Explanation: Let $x = 0.\overline{45}$. Multiply by 100 to shift the two repeating digits, then subtract the original equation:

$$\begin{array}{rl} 100x = & 45.\overline{45} \\ -\quad x = & 0.\overline{45} \\ \hline 99x = & 45 \implies \end{array}$$

$$x = \frac{45}{99} = \frac{45 \div 9}{99 \div 9} = \frac{5}{11}.$$

6. $1.\overline{27}$

   Explanation: Divide 14 by 11 using the bus stop method:

$$\begin{array}{r|l} & 1.\ 2\ 7\ 2\ 7\ldots \\ \hline 11 & 14.^{3}0^{8}0^{3}0^{8}0\ldots \end{array}$$

   The remainders 3 and 8 alternate indefinitely, giving $\dfrac{14}{11} = 1.\overline{27}$.

7. $\dfrac{7}{8}$ **inches**

   Explanation: Write 0.875 as a fraction over 1000 and simplify by dividing by 125:

$$0.875 = \frac{875}{1000} = \frac{875 \div 125}{1000 \div 125} = \frac{7}{8} \text{ inches.}$$

8. 0.3125 **liters**

   Explanation: Divide 5 by 16 using the bus stop method:

$$\begin{array}{r|l} & 0.\ 3\ 1\ 2\ 5 \\ \hline 16 & 5.^{5}0^{2}0^{4}0^{8}0 \end{array}$$

   Thus, $\dfrac{5}{16} = 0.3125$ liters.

9. $-2\dfrac{9}{25}$ **dollars**

   Explanation: Retain the negative sign and simplify the fractional part $\dfrac{36}{100}$ by dividing by 4:

$$-2.36 = -\left(2 + \frac{36}{100}\right) = -\left(2 + \frac{36 \div 4}{100 \div 4}\right) = -2\frac{9}{25} \text{ dollars.}$$

10. 11.15 **seconds**

    Explanation: Convert the fractional part $\dfrac{3}{20}$ to hundredths and add to the whole number:

$$11\frac{3}{20} = 11 + \frac{3 \times 5}{20 \times 5} = 11 + \frac{15}{100} = 11.15 \text{ seconds.}$$